

KodiKit Operations Guide

Operating a self-updating Kodi add-on repository, and provisioning a Fire TV Stick 4K from it.

Two procedures and a worksheet. **Book One** sets up GitHub and runs the repository — the thing that keeps add-ons updating by themselves. **Book Two** takes a Fire TV Stick from factory state to a finished box in 34 numbered steps. **Book Three** is a worksheet to fill in as you go.

Start at Book One, Part 1 if GitHub has never been set up on this machine. If the repository is already published, you only need Book Two.

Everything is written as steps to follow in order. Each step is one command to type or one control to click, and lines marked **Check** tell you what success looks like — if you do not see it, stop there rather than continuing. Background, file maps and troubleshooting live in the appendices so the procedures stay short.

Two files in the repository root pair with these instructions: `addons.template.json` holds the three add-on entry shapes ready to paste, and `repo.config.json` is the identity you set in Part 2.

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BOOK ONE

GitHub setup and running the repository

Every section here is a numbered procedure. Run the steps in order; each one is a command to type or a control to click. Lines marked **Check** tell you what success looks like — if you do not see it, stop there rather than continuing.

Start at Part 1 if you have never set up GitHub on this machine. If `gh auth status` already reports you as logged in, skip to Part 2.

Background and troubleshooting live in the appendices, so the procedures stay short.

Part 1 — GitHub setup

Do this once per machine.

1.1 Create a GitHub account — skip if you have one.

Go to github.com/signup and register. Your username becomes part of every URL your devices fetch from, so pick one you are willing to keep.

1.2 Install the tools.

macOS (Homebrew):

```
brew install git gh
```

Debian / Ubuntu:

```
sudo apt update && sudo apt install git gh python3
```

Windows:

```
winget install Git.Git GitHub.cli Python.Python.3.12
```

Check: all three report a version.

```
python3 --version      # need 3.8 or newer
git --version
gh --version
```

1.3 Tell git who you are. Commits fail or land under the wrong name without this.

```
git config --global user.name "Your Name"
git config --global user.email "you@example.com"
```

Check:

```
git config --global user.name && git config --global user.email
```

Use the email attached to your GitHub account, or commits will not be linked to your profile. GitHub also offers a `@users.noreply.github.com` address if you would rather not publish a real one.

1.4 Sign in to GitHub from the terminal.

```
gh auth login
```

It asks four questions. The answers:

Prompt	Answer
What account do you want to log into?	GitHub.com
What is your preferred protocol for Git operations?	HTTPS
Authenticate Git with your GitHub credentials?	Yes
How would you like to authenticate?	Login with a web browser

It then shows a one-time code. Copy it, press Enter, and your browser opens; paste the code and approve. The token is stored in your system credential store.

Check:

```
gh auth status
```

You want Logged in to github.com account YOURNAME and a token with the repo, read:org and gist scopes.

Headless machine, or no browser? Create a personal access token (classic) at github.com/settings/tokens with the repo, read:org and gist scopes, then pipe it in: `gh auth login --with-token < token.txt`.

1.5 Get the repository onto this machine.

If you already have the directory, just note where it is. Otherwise clone it:

```
git clone https://github.com/YOURNAME/YOURREPO.git ~/Dev/kodi-repo
```

Starting from scratch instead? Copy the repository files into a new directory and:

```
cd ~/Dev/kodi-repo && git init -b main
```

Part 2 — Configure the repository

2.1 Open the repository directory.

```
cd ~/Dev/kodi-repo
```

2.2 Set your identity. Open `repo.config.json` and set these four fields. Record them in the worksheet (Book Three) as you go — the install URL in Part 3 is built from them.

Field	Set it to
<code>github_user</code>	Your GitHub username or org, exactly as it appears in URLs
<code>github_repo</code>	The repository name you will push to
<code>repo_name</code>	The display name shown in Kodi. Anything readable.
<code>provider</code>	Your name, shown as the author

Leave `repo_id`, `branch`, `hosting` and `keep_versions` alone unless Part 7 applies.

2.3 Build.

```
python3 scripts/build_repo.py
```

Check: the last two lines report the add-on count, an md5, and the URL it will serve from. That URL must contain your `github_user` and `github_repo` from 2.2.

2.4 Check dependencies.

```
python3 scripts/check_deps.py
```

Check: all dependencies resolve. If not, go to Appendix C.

2.5 Commit.

```
git add -A && git commit -m "Configure repository identity"
```

Part 3 — Publish to GitHub

Do this once. After it, everything is automatic.

3.1 Create the repository. It must be public — Kodi cannot authenticate.

```
gh repo create YOURNAME/YOURREPO --public --source=. --remote=origin
```

If that reports `remote origin already exists`, the directory is already pointed somewhere. Check with `git remote -v`, then either use that target or repoint it: `git remote set-url origin URL`.

3.2 Push.

```
git push -u origin main
```

Check: no authentication prompt. If you get one, redo 1.4.

3.3 Wait for the build.

```
gh run watch
```

Check: the run completes green. If it fails, read the log — it is running the same two commands you ran in 2.3 and 2.4.

3.4 Verify what Kodi will see. Replace `YOURNAME` and `YOURREPO` in the first line, then paste the whole block.

```
BASE=https://raw.githubusercontent.com/YOURNAME/YOURREPO/main/docs
curl -sI $BASE/addons.xml | head -1          # expect: HTTP/2 200
curl -s $BASE/addons.xml.md5                # expect: 32 hex characters, nothing else
curl -sI $BASE/repository.kodikit.zip | head -1  # expect: HTTP/2 200
```

Check: all three as described. A 404 on the first two means 2.2 does not match where you actually pushed.

3.5 Enable the scheduled sync. Open the repository on github.com → **Actions** tab → if prompted, click **I understand my workflows, go ahead and enable them**.

Check: the **Sync** workflow is listed and not disabled.

3.6 Write down the install URL. This is what you type into Kodi in Book Two, step 17:

```
https://raw.githubusercontent.com/YOURNAME/YOURREPO/main/docs
```

No gh? Create the repository at github.com/new — public, no README, no .gitignore — then: `git remote add origin https://github.com/YOURNAME/YOURREPO.git` && `git push -u origin main`. When git asks for a password, paste a personal access token; GitHub stopped accepting account passwords for this in 2021.

Part 4 — Add an add-on

The repeatable procedure. Six steps.

4.1 Find the add-on's ID. Not its display name. Any one of:

Source	Where to look
Its GitHub repo	<code>addon.xml</code> → the <code>id=</code> attribute of the root <code><addon></code> element
An installed copy	The folder name under <code>~/.kodi/addons/</code> , e.g. <code>script.trakt/</code>
Kodi	Add-ons → the add-on → Information

4.2 Decide which section it goes in. Run this, with your IDs in the `probe` list:

```
python3 - <<'EOF'
import gzip, re, urllib.request
url = "https://mirrors.kodi.tv/addons/omega/addons.xml.gz"
req = urllib.request.Request(url, headers={"User-Agent": "kodikit"})
raw = gzip.decompress(urllib.request.urlopen(req, timeout=60).read()).decode("utf-8",
"replace")
ids = set(re.findall(r'<addon\s+id="([^\"]+)"', raw))
for probe in ["script.trakt", "service.vpn.manager"]:
    print("%-34s %s" % (probe, "IN official" if probe in ids else "NOT in official"))
EOF
```

Result	Section	Why
IN official	official	Installed by ID; stays on the maintainer's update channel
NOT in official	mirror	Packaged into your repo from its GitHub source
Installs only from a vendor's own repo	external	Documented, never auto-installed — see Appendix A

4.3 Add the entry to `addons.json`. Copy the matching block from `addons.template.json` in the repository root — it holds all three shapes ready to paste, so you do not have to retype them from this page. The same shapes are reproduced below for reference.

For `mirror`:

```
{
  "id": "service.vpn.manager",
  "name": "VPN Manager for OpenVPN",
  "category": "privacy",
  "why": "One sentence on what it does and why it earned a slot.",
  "github": "Zomboided/service.vpn.manager",
  "track": "release",
  "optional": true
}
```


For `official` — no `github`, no `track`:

```
{
  "id": "script.trakt",
  "name": "Trakt",
  "category": "tracking",
  "why": "Scrobbling, watched-state sync and lists across devices."
}
```

For `external` — different shape entirely: no `id`, a `vendor url`, and the add-ons it provides:

```
{
  "name": "TheMovieDb Helper (6.x)",
  "why": "Hard-requires script.module.pil, which no official repo ships.",
  "url": "https://jurialmunkey.github.io/repository.jurialmunkey/",
  "addons": ["plugin.video.themoviedb.helper", "script.skinvariables"]
}
```

Field reference:

Field	Sections	Meaning
<code>id</code>	mirror, official	Must match the add-on's own <code>addon.xml</code> exactly
<code>name</code>	all	Display name in the Toolbox picker
<code>category</code>	mirror, official	Groups and sorts the picker. In use: subtitles, playback, tracking, tools, interface, privacy.
<code>why</code>	all	Your justification. Ends up in the README.
<code>github</code>	mirror	owner/repo
<code>track</code>	mirror	release (newest release, falling back to branch head) or branch
<code>ref</code>	mirror	Pin a branch, e.g. <code>"ref": "matrix"</code>
<code>optional</code>	mirror, official	true leaves it unticked in the picker

4.4 Bump the Toolbox version. Open `src/script.kodikit.toolbox/addon.xml` and increase the patch number, e.g. `1.2.1` → `1.2.2`.

Do not skip this. The curated list ships *inside* the Toolbox. Change the list without changing the version and the published index stays byte-identical, so no device ever fetches your new list — while the build looks completely successful. `build_repo.py` refuses to finish in that state and names the stale add-on.

4.5 Build and check.

```
python3 scripts/build_repo.py && python3 scripts/check_deps.py
```

Check: exit code 0, and all dependencies resolve. If the build stops with content changed without a version bump, you missed 4.4.

4.6 Push.

```
git add -A && git commit -m "Add <addon-id>" && git push
```

Check: run the verification in Part 9.

Part 5 — Remove an add-on

5.1 Delete its entry from `addons.json`.

5.2 Bump the Toolbox version in `src/script.kodikit.toolbox/addon.xml`.

5.3 Build and push:

```
python3 scripts/build_repo.py && python3 scripts/check_deps.py
git add -A && git commit -m "Remove <addon-id>" && git push
```

Check: `grep '<addon id=<addon-id>'' docs/addons.xml` returns nothing.

Copies already installed on devices keep working but stop receiving updates. A repository cannot retract an add-on — to remove it from a device, uninstall it there.

Part 6 — Change the Toolbox itself

6.1 Edit the code under `src/script.kodikit.toolbox/`.

6.2 Bump the version in its `addon.xml` — patch for a fix, minor for new behaviour.

6.3 Build, check, push:

```
python3 scripts/build_repo.py && python3 scripts/check_deps.py
git add -A && git commit -m "Toolbox: <what changed>" && git push
```

Part 7 — Change the repository's identity or URLs

Only when moving the repo or renaming it.

7.1 Edit `repo.config.json`: `github_user`, `github_repo`, `branch`, or `hosting`.

7.2 Bump `repo_version` in the same file.

7.3 Build and push as in 6.3.

7.4 Reinstall the repository zip on **every device**. The old URLs are baked into the copy they already have.

Because of 7.4, get `github_user` and `github_repo` right in Part 2 and leave them alone.

Part 8 — Daily operation

Nothing to do. For reference, the scheduled job at 05:15 UTC:

1. Checks each mirrored add-on's upstream for a newer release.
2. Repackages anything that moved.
3. Runs the dependency check; a failure stops the commit.
4. Commits the regenerated docs/ only if something changed.

Devices poll roughly every 24 hours, so worst case is about 48 hours from an upstream release to your stick.

To force a run now:

```
gh workflow run sync.yml && gh run watch
```

Part 9 — Health check

Run after any push, or whenever something looks wrong.

9.1 The index lists what you expect.

```
grep -o 'id="[^\"]*" name="[^\"]*" version="[^\"]*" docs/addons.xml
```

9.2 A specific add-on is present.

```
grep -o 'id="service.vpn.manager"[^>]*version="[^\"]*" docs/addons.xml
```

9.3 Its zip is Kodi-shaped — `<id>/addon.xml` must be at the root:

```
unzip -l docs/zips/service.vpn.manager/*.zip | head -5
```

9.4 The Toolbox carries the current list.

```
python3 -c "import zipfile,json,glob; z=sorted(glob.glob('docs/zips/script.kodikit.toolbox/*.zip'))[-1]; \
m=json.loads(zipfile.ZipFile(z).read('script.kodikit.toolbox/resources/addons.json')); \
print(sorted(e['id'] for s in ('mirror','official') for e in m[s]))"
```

9.5 The live URLs still respond. Re-run 3.4.

9.6 On a device: Toolbox → Install curated add-ons. New entries appear in the picker, unticked if you marked them `optional`.

Appendix A — How it works

The update loop. Kodi downloads `addons.xml.md5` (32 bytes) on a ~24 hour cycle. If that hash changed, it downloads `addons.xml`, compares every `version=` against what is installed, and downloads anything newer from `docs/zips/`. So an update happens **only** when a version number increases. Re-uploading a zip at the same version does nothing; lowering a version does nothing. A broken release is fixed with a higher patch version, never a re-upload.

The three routes. `official` keeps an add-on on its maintainer's update channel — mirroring something Kodi already ships would fork it onto your channel and make you responsible for tracking it. `mirror` is for add-ons genuinely unavailable elsewhere. `external` is for add-ons whose dependencies nobody publishes: TheMovieDb Helper 6.x hard-requires `script.module.pil`, a compiled Pillow build no official repo ships, so mirroring it would deliver an add-on that cannot install.

What the builder does. Syncs mirrors from upstream; packages everything in `src/` at the version in its `addon.xml`; generates the repository add-on from `repo.config.json`; regenerates the index, checksum, `index.html`, and prunes old zips. Zips use fixed timestamps, so identical content produces identical bytes — otherwise CI would commit noise daily.

What the dependency checker does. Fetches Kodi's official index for Omega and Piers, unions it with your own IDs, and verifies every non-optional `<import>` resolves. It fails the build rather than letting a device discover the problem.

Where the add-on ID appears. Folder name in `docs/zips/`, zip filename, folder at the root inside the zip, and the `id=` in `addon.xml`. Kodi builds its download URL from the ID and version in the index, so any mismatch is a 404 that surfaces as "failed to install". The builder derives all four from one source, which is why you never rename zips by hand.

Appendix B — File map

Path	Purpose	Edit?
<code>addons.json</code>	The curated list	Yes — this is the control surface
<code>addons.template.json</code>	Copy-paste entry templates; never read by the build	Reference
<code>repo.config.json</code>	Repo identity, hosting, versions	Yes
<code>src/</code>	Add-ons you maintain	Yes
<code>scripts/build_repo.py</code>	The builder	Rarely
<code>scripts/check_deps.py</code>	Dependency validation	Rarely
<code>scripts/make_art.py</code>	Regenerates icon/fanart (needs Pillow)	Rarely
<code>assets/</code>	Committed icon and fanart	Rarely
<code>docs/</code>	Published tree — what Kodi downloads	Never by hand
<code>guides/</code>	This document's source	Yes

Appendix C — Troubleshooting

Symptom	Cause	Fix
<code>gh auth login</code> opens no browser	Headless machine	Use the token method in the 1.4 note
<code>git push</code> asks for a password	Not authenticated, or using a password	Redo 1.4, or paste a personal access token
Commits show the wrong author	<code>user.email</code> not your GitHub address	Redo 1.3

Symptom	Cause	Fix
Build stops: content changed without a version bump	The working case, caught	Bump the version in <code>src/<addon>/addon.xml</code> , rebuild
check_deps reports an unresolvable import	A dependency nobody publishes	Move the add-on to external and document the vendor repo
Mirror sync: no published release, falling back to branch head	Upstream publishes no releases	Normal. Set <code>"track": "branch"</code> to make it explicit.
Mirror sync: add-on not found in the tarball	id does not match upstream's <code>addon.xml</code>	Correct the id
Mirror sync fails after an upstream rename	Default branch changed	Set <code>"ref"</code> explicitly
Actions never run	Workflows not enabled on a new repo	Do 3.5
Kodi: "Could not connect to repository"	<code>github_user/github_repo</code> wrong, or repo private	Re-run 3.4; confirm the repo is public
Kodi: repository installs but lists nothing	Index empty, or checksum stale	Re-run 2.3; never hand-edit docs/
Kodi: "Failed to install add-on"	Zip name does not match <code>id-version.zip</code>	Re-run 2.3; the builder names them
Kodi: "Dependency not met"	Unresolvable <code><import></code>	Run 2.4
An add-on never updates on one device	Per-add-on auto-update set to Never	Add-on info → Auto-update → On
Updates appear only after a restart	Normal 24-hour poll cycle	Force a check on the device, or wait

BOOK TWO

Provisioning a Fire TV Stick, step by step

One continuous procedure, steps 1 to 34, taking a Fire TV Stick from factory state to a finished, self-updating Kodi box. Target hardware: **Fire TV Stick 4K (2018, model E9L29Y)** — 1.5 GB RAM, 8 GB storage, Android 7.1+, Kodi 21.1. Notes for the 1 GB gen 1/2 sticks appear where they differ.

Lines marked **Check** tell you what success looks like. Roughly 30 minutes end to end:

Stage	Steps	Time
A Prepare the stick	1–5	3 min
B Install Kodi	6–8	5 min
C Surfshark	9–13	5 min
D Repository	14–20	5 min
E Add-ons	21–23	5 min
F Configure	24–30	5 min
G Verify	31–34	3 min

Doing Stage A step 3 (ADB) costs a minute and saves more than that later — steps 17, 32 and 33 all use it.

Before you start, have these ready:

- The install URL from Book One, step 3.6
- Your Surfshark login
- The Kodi armeabi-v7a APK for your version, or the Downloader app
- Optional but recommended: ADB on your computer, for the verification steps

Stage A — Prepare the stick (steps 1–5)

1. On the stick: Settings → My Fire TV → About → click **Fire TV Stick** seven times to reveal Developer Options.
2. Settings → My Fire TV → Developer Options → **Apps from Unknown Sources** → **On**.
3. (Optional, for the verification steps) Same menu → **ADB debugging** → **On**.
4. Settings → My Fire TV → About → Network → note the **IP address**.
5. (Optional) From your computer:

```
adb connect STICK-IP:5555
```

Check: connected to STICK-IP:5555. Accept the prompt on the TV if one appears.

Stage B — Install Kodi (steps 6–8)

6. Install Kodi. Pick one route.

Route A — Downloader app on the stick. Open Downloader, type this exact URL, press Go, then Install when it finishes:

```
mirrors.kodi.tv/releases/android/arm/kodi-21.1-0mega-armeabi-v7a.apk
```

Route B — ADB from your computer. Download the same file, then:

```
adb install -r kodi-21.1-0mega-armeabi-v7a.apk
```

Two things to get right. Use **armeabi-v7a**, never arm64-v8a — the stick is 32-bit ARM and the wrong file simply fails to install. And if Kodi is already on the stick, install the **same version it already runs**, or its profile may not survive the change. Newer builds live in the same directory (21.3 is current at time of writing); browse mirrors.kodi.tv/releases/android/arm/ to pick one.

7. Launch Kodi once, let it reach the home screen, then quit it. This creates the profile directory.
8. Skip Kodi's setup wizard if offered. Leave the skin as Estuary.

Stage C — Set up Surfshark (steps 9–13)

Do this **before** any sign-in inside Kodi. Sign-ins bind to the address that completed them, so doing this later means redoing them.

9. On the stick: Amazon appstore → search **Surfshark** → install.
10. Open Surfshark, sign in, connect to a location.

11. In Surfshark's settings, enable **auto-connect on startup**, so a reboot cannot silently drop protection.
12. (Optional) Set **Bypasser** if you want specific apps outside the tunnel.
13. Verify the tunnel at OS level — the app's own UI is not proof. (Needs ADB from step 5.)

```
adb shell ip addr show tun0
```

Check: a `tun0` interface exists with an IP. If it does not, the stick is not tunnelling no matter what the app displays.

Do not use the VPN Manager add-on on this device. It raises an OpenVPN tunnel from inside Kodi, which an ordinary Android app cannot do on Fire OS. It also bundles profiles for NordVPN, ExpressVPN and PIA only — not Surfshark. It stays in the repository, unticked, for Linux and LibreELEC boxes; Appendix C covers using it with Surfshark there.

Stage D — Install the repository (steps 14–20)

14. Kodi → Settings (gear) → System → Add-ons → **Unknown sources** → **On**. Accept the warning.

15. Settings → Add-ons → Updates → **Install updates automatically**.

Step 15 goes before any add-on exists so every one inherits it. This single setting is what makes the stick maintenance-free.

16. Settings → File manager → **Add source** → `<None>`.

17. Enter the install URL from Book One, step 3.6:

```
https://raw.githubusercontent.com/YOURNAME/YOURREPO/main/docs
```

Do not type this on the remote. It is the slowest minute of the whole procedure and a single wrong character fails silently at step 19. With ADB connected (step 5), tap the text field once so the cursor is in it, then type it from your computer:

>

```
adb shell input text "https://raw.githubusercontent.com/YOURNAME/YOURREPO/main/docs"
```

>

The text appears in the field. Check it on screen before pressing OK.

18. Name it `kodikit` → OK.

19. Kodi → Add-ons → **Install from zip file** → `kodikit` → `repository.kodikit.zip`.

Check: a notification confirms the repository add-on installed.

20. Add-ons → **Install from repository** → KodiKit → Program add-ons → **KodiKit Toolbox** → Install.

Check: the Toolbox appears under Add-ons → Program add-ons.

If step 19 fails, the URL in step 17 is wrong. Re-run Book One step 3.4 from your computer.

Stage E — Install the add-ons (steps 21–23)

21. Open **KodiKit Toolbox** → **Install curated add-ons**.

22. In the picker, untick anything you do not want. For this stick:

Untick	Reason
<code>skin.copacetic</code>	40–150 MB and slower menus; Estuary is lighter
<code>script.plexmod</code>	Only useful with a Plex server
<code>pvr.iptvsimple</code>	Only useful with an IPTV M3U subscription
<code>service.vpn.manager</code>	Wrong route on Fire OS — Stage C already covered it
<code>script.service.janitor</code>	Only if the stick stores local recordings

Leave ticked: InputStream Adaptive and FFmpeg Direct, a4kSubtitles, Black Bars Never, Up Next, Logfile Uploader, YouTube.

23. Confirm and let it run.

Check: the Toolbox reports what installed. Anything already present is skipped automatically.

Stage F — Configure (steps 24–30)

24. Toolbox → **Apply streaming cache tuning** → select **Fire TV Stick** → Write → **Restart** when prompted.

Do not pick *Balanced* here. Its 64 MB buffer becomes roughly 192 MB resident, which does not fit alongside Fire OS on 1.5 GB. On a 1 GB gen 1/2 stick, pick *Low memory* instead.

25. Settings → Player → Language → **Default TV show service** and **Default movie service** → set both to **a4kSubtitles**.

Subtitles do nothing until this is set. It is the single most common "it's broken" report.

26. Settings → Player → Videos → **Allow hardware acceleration – MediaCodec (Surface)** → **On**.

27. Same screen → **Allow hardware acceleration – MediaCodec** → **Off** (redundant with Surface, costs memory).

28. Same screen → **Adjust display refresh rate** → **On start/stop**; **Sync playback to display** → **Off**; **Enable HQ scalars** → **0%**.

29. Settings → Interface → **Show RSS news feeds** → **Off**. Settings → Media → General → **Show media flags** → **Off**.

30. Settings → Services → Control → **Allow remote control via HTTP** → **On**, port 8080.

Step 30 is what makes remote diagnosis possible later without touching the device.

Stage G — Verify (steps 31–34)

31. Reboot the stick, then leave Kodi sitting on the home screen for a minute.

32. Check memory and disk from your computer. (Needs ADB from step 5. Without it, use Toolbox → **Storage report** for the disk figure; there is no on-device view of memory.)

```
adb shell dumpsys meminfo org.xbmc.kodi | grep 'TOTAL PSS'
adb shell du -sh /sdcard/Android/data/org.xbmc.kodi/files/.kodi
```

Check: against the targets below.

Metric	Stick 4K	1 GB sticks
TOTAL PSS idle	under 600 MB	under 420 MB
.kodi/ on disk	under 500 MB	under 350 MB
Cold start	under 18 s	under 25 s
Background services	6 or fewer	4 or fewer

33. Confirm auto-update really is set:

```
adb shell "grep addonupdates \
/sdcard/Android/data/org.xbmc.kodi/files/.kodi/userdata/guisettings.xml"
```

Check: the value is 0. If it is 1 or 2, redo step 15.

34. Play something for a minute and confirm it is smooth — no stutter, no audio drifting out of sync. With a USB or Bluetooth keyboard attached, **Ctrl+Shift+0** overlays the codec panel with a dropped-frame counter, which is the precise version of the same check.

Done. The stick now updates itself. Every add-on tracks its own upstream through the repository, with no further action.

After setup — routine maintenance

When	Do
Monthly	Toolbox → Storage report
When storage looks bad	Toolbox → Clean up caches
Occasionally	Toolbox → Clean video/music library
After adding an add-on	Re-run step 32 to catch a heavy background service

Artwork re-downloads as you browse after a cache clean. That is expected; the first few minutes feel slower.

Troubleshooting

Symptom	Fix
Kodi APK will not install	Wrong architecture — use <code>armeabi-v7a</code>
Step 19 fails	URL in step 17 is wrong; re-run Book One step 3.4
Repository installs but lists nothing	Book One, Appendix C
Subtitles never appear	Step 25 not done
Buffering on 1080p or 4K	Redo step 24 and pick Fire TV Stick
Stutter on high-bitrate video	Step 26 — MediaCodec (Surface) is off
Black screen with audio	Step 27 — disable the non-Surface MediaCodec option
Very slow menus	A third-party skin got installed; switch back to Estuary
Out of space within weeks	Toolbox → Clean up caches; confirm step 24 was applied
Add-ons never update	Step 33 returns something other than 0
Surfshark shows connected but traffic is not tunnelled	Step 13 shows no <code>tun0</code> ; reconnect in the app
Everything is slow on a gen 1/2 stick	1 GHz dual-core is the limit — cut the add-on count, expect 1080p only

Appendix A — The hardware

	Stick gen 1 (2014)	Stick gen 2 (2016)	Stick 4K (2018)	Stick Lite / 3rd gen
Model	W87CUN	LY73PR	E9L29Y	—
RAM	1 GB	1 GB	1.5 GB	1 GB
Storage	8 GB (~5 usable)	8 GB (~5 usable)	8 GB (~5 usable)	8 GB
CPU	Dual-core 1.0 GHz	Quad-core 1.3 GHz	Quad-core 1.7 GHz	Quad-core 1.7 GHz
Android base	5.1	5.1	7.1+	9
Realistic Kodi ceiling	19–20	20	21+	21+

Fire OS reserves 350–450 MB. Kodi with Estuary is 250–300 MB. That leaves roughly 250 MB of headroom on a 1 GB stick and 700 MB on the 4K.

Storage, not RAM, is the binding constraint on the 4K. The extra 512 MB buys room for services, but the flash is the same 8 GB and the texture cache grows without limit until something caps it — which is what step 24 does.

On gen 1/2 sticks (Android 5.1) the practical Kodi ceiling is 20; check kodi.tv/download for the current `armeabi-v7a` minimum, as it rises with each release.

Appendix B — What step 24 writes

Toolbox → cache tuning → Fire TV Stick produces `userdata/advancedsettings.xml`, backing up any existing file with a timestamp first:

```
<?xml version="1.0" encoding="UTF-8" standalone="yes"?>
<advancedsettings>
  <cache>
    <buffermode>1</buffermode>
    <memorysize>41943040</memorysize>
    <readfactor>4</readfactor>
  </cache>
  <gui>
    <algorithmdirtyregions>3</algorithmdirtyregions>
  </gui>
  <imageres>540</imageres>
  <fanartres>720</fanartres>
  <network>
    <curlclienttimeout>20</curlclienttimeout>
    <curlallowspendtime>15</curlallowspendtime>
  </network>
</advancedsettings>
```


- `memorysize` **40 MB**. Kodi allocates roughly three times the configured value, so this is ~120 MB resident. *Low memory* (20 MB → ~60 MB) suits 1 GB sticks; *Balanced* (64 MB → ~192 MB) needs a box with real RAM.
- `imageres / fanartres`. Dropping fanart from 1080 to 720 more than halves texture cache growth, and on a TV that upscales anyway the difference is invisible. This is the most effective single setting against the 8 GB flash filling.
- `algorithmdirtyregions` **3**. Redraw only what changed.
- **Network timeouts**. A dead source fails instead of hanging the UI.

Cache settings take effect only after the restart in step 24.

Appendix C — VPN Manager with Surfshark, on a non-Android box

Only relevant on Linux, LibreELEC or Windows, where an add-on can execute OpenVPN.

1. Toolbox → Install curated add-ons → tick **VPN Manager for OpenVPN**.
2. VPN Manager → settings → provider → **User Defined**.
3. Download Surfshark's `.ovpn` config files from their manual-setup page.
4. Import them with VPN Manager's import wizard.
5. Authenticate with Surfshark's **manual-setup service credentials** from your account dashboard — a separate username and password, **not** your Surfshark login.

Step 5 is the usual failure: the account email and password do not work for manual OpenVPN.

Appendix D — Why there is no snapshot build

The original goal was a "build" — a packaged `userdata` + `addons` snapshot installed by a wizard. That shape is wrong here for four reasons:

1. **It cannot update itself**. Add-ons inside a build are sideloaded, belong to no repository, and never receive updates.
2. **Reinstalling wipes your configuration**, because updating means replacing `userdata`.
3. **Builds rot across Kodi releases** — settings move and the add-on database schema changes.
4. **Builds leak credentials**. `userdata/addon_data/` holds API keys and account logins; packaging it ships your accounts to everyone who installs it.

Stages D–F deliver what people actually want from a build — one action, whole set installed, sensible settings — while every add-on stays individually updatable.

If you genuinely need one (cloning many identical sticks, or no reliable network):

```
cd ~/.kodi
zip -qr ~/mybuild-1.0.0.zip addons userdata \
  -x 'addons/packages/*' 'userdata/Database/*' 'userdata/Thumbnails/*' \
```

```
'*/__pycache__/*' '*.pyc' '*.log' 'temp/*'  
grep -rile 'token|apikey|password|secret' ~/.kodi/userdata/addon_data/
```

Review whatever that `grep` lists before sharing the zip. Restore by extracting over `.kodi/` with Kodi **force-stopped** — it rewrites `guisettings.xml` on exit and will otherwise overwrite what you restored:

```
adb shell am force-stop org.xbmc.kodi  
adb push mybuild-1.0.0.zip /sdcard/Download/  
adb shell "cd /sdcard/Android/data/org.xbmc.kodi/files/.kodi && \  
  unzip -o /sdcard/Download/mybuild-1.0.0.zip"
```

Install the repository afterwards so the cloned devices resume receiving updates.

BOOK THREE

Worksheet

Worksheet

Fill this in as you follow the guide. Every blank is used by a numbered step, and the step is named beside it.

1. Your values

Fill these in once, in Book One, Part 1.

What	Used by	Your value
GitHub username or org	2.2 github_user	
Repository name	2.2 github_repo	
Repository display name	2.2 repo_name	
Provider / author name	2.2 provider	
Install URL	3.6, and Book Two step 17	https://raw.githubusercontent.com/_____/_____/main/docs
Fire TV Stick IP address	Book Two steps 4, 5	
Kodi version on the stick	Book Two step 6	

2. Add-ons you want

One row per add-on. Fill the first two columns, run the probe in step 4.2, then complete the rest.

Display name	Add-on ID	Probe result	Section	Optional?	Added
		IN / NOT in official	mirror / official / external	yes / no	[]
					[]
					[]
					[]
					[]
					[]
					[]
					[]

Reminder: after adding entries, **bump the Toolbox version** (step 4.4) or no device will ever see them.

3. Book One checklist

Part 1 — GitHub setup (*once per machine; skip if `gh auth status` already shows you logged in*)

- [] 1.1 GitHub account exists
- [] 1.2 `git`, `gh` and `python3` installed, all reporting a version
- [] 1.3 `git config --global user.name` and `user.email` set
- [] 1.4 `gh auth login` done; `gh auth status` shows *Logged in*
- [] 1.5 Repository directory on this machine

Part 2 — Configure the repository

- [] 2.1 In the repository directory
- [] 2.2 Four identity fields set in `repo.config.json`
- [] 2.3 `build_repo.py` ran; served-from URL matches your values
- [] 2.4 `check_deps.py` says *all dependencies resolve*
- [] 2.5 Committed

Part 3 — Publish

- [] 3.1 Repository created, public
- [] 3.2 Pushed, with no authentication prompt
- [] 3.3 Actions run green
- [] 3.4 All three URLs verified (200 / 32-char hash / 200)
- [] 3.5 Workflows enabled in the Actions tab
- [] 3.6 Install URL written into section 1 above

4. Book Two checklist

- [] **Stage A** (1–5) Developer options on, unknown sources on, IP noted
- [] **Stage B** (6–8) Kodi installed (`armeabi-v7a`), launched once, quit
- [] **Stage C** (9–13) Surfshark installed, auto-connect on, `tun0` confirmed
- [] **Stage D** (14–20) Auto-update set *before* add-ons, source added, repository and Toolbox installed
- [] **Stage E** (21–23) Curated add-ons installed, unwanted entries unticked
- [] **Stage F** (24–30) Cache profile applied, subtitle default set, MediaCodec Surface on
- [] **Stage G** (31–34) Rebooted, memory and disk within targets, `addonupdates = 0`, playback clean

5. Measured results

From Book Two, step 32. Compare against the targets in the same step.

Metric	Target (Stick 4K)	Measured
TOTAL PSS idle	under 600 MB	
.kodi/ on disk	under 500 MB	
Cold start	under 18 s	
Background services	6 or fewer	
general.addonupdates	0	